

PREHISTORICAL VR SIMULATION AS AN EDUCATIONAL CONTENT

PROJECT PROPOSAL REPORT

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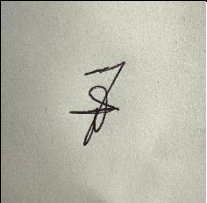
**BSc (Hons) Degree in Information Technology Specialized in Interactive
Media**

Sri Lanka Institute of Information Technology

2025

Declaration of Candidate & Supervisor

I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text

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Date

Abstract

It was the means through which Virtual Reality made education revolutionary, being able to dive into learning virtually. Contemporary educational uses of VR barely scratch the surface in making deep simulation of prehistoric extinction events-as much as volcanic eruptions or meteorite impacts-limits comprehension of their effects on the environment and biosphere. This research proposes an AI-driven, interactive VR simulation that enables the user to experience and manipulate prehistoric events in real-time for a better comprehension of the dynamics of extinction and evolution. The system will be characterized by scientifically correct 3D environments, interactive extinction scenarios, and real-time feedback, enhancing user engagement and knowledge retention. It represents a large, important research gap in educational VR applications, ranging from user-triggered events to adaptive simulations, with the development of an innovative, open platform for studying Earth's history and biodiversity based on experiential learning.

Keywords: Virtual Reality, Prehistoric Extinction, Interactive Learning, Immersive Education, AI-Driven Simulations, Environmental Dynamics, Knowledge Retention, User Engagement.

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1. Introduction

1.1 Background Literature

Virtual Reality (VR) has emerged as a powerful tool in education, providing immersive learning experiences that traditional methods fail to achieve. Studies indicate that VR-based learning can increase knowledge retention by 75% compared to traditional techniques due to its ability to provide interactive, engaging, and realistic simulations of complex concepts . VR enables learners to visualize abstract ideas, explore environments otherwise inaccessible, and actively participate in simulations, enhancing both comprehension and retention.

The application of VR in prehistoric studies, particularly focusing on extinction events caused by volcanic eruptions and meteorite impacts, remains underexplored . Traditional educational resources such as books, documentaries, and museum exhibits primarily offer static representations of prehistoric events. These methods often fall short of delivering an interactive and experiential understanding of the factors contributing to extinction events, such as the environmental changes triggered by volcanic eruptions and meteorite impacts.

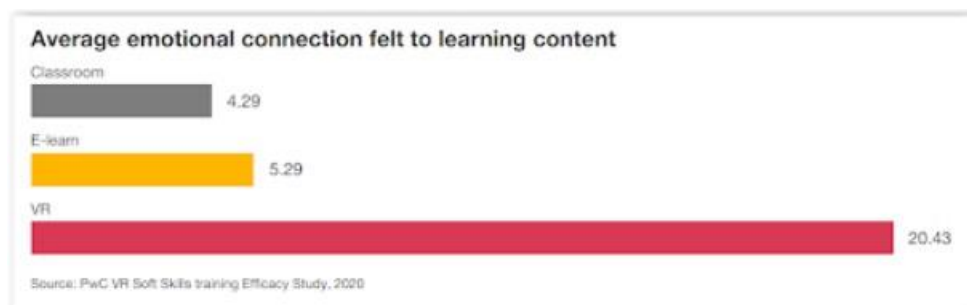


Fig 01. Average emotional connection felt to learning content using VR By students

1.2 Research Gap

Despite the rapid growth of VR applications in education, prehistoric extinction events remain underexplored. Most existing VR simulations focus on visualizing historical or natural phenomena but lack interactivity, limiting users' ability to engage with cause-and-effect relationships in real-time. Current VR experiences of events like volcanic eruptions or meteorite impacts are largely static, showing predefined scenarios without allowing users to manipulate key variables such as eruption magnitude or ash dispersion. This restricts the potential of VR as an immersive educational tool.

An AI-driven approach could significantly enhance these simulations by dynamically modeling environmental changes, geological shifts, and evolutionary responses to extinction events. AI could enable real-time interactivity, allowing users to influence events and witness their long-term consequences. For instance, ecosystems within the simulation could evolve based on user actions, demonstrating how flora and fauna adapt—or perish—due to shifting climates or catastrophic disruptions.

Such AI-integrated VR experiences would transform traditional VR from a passive visualization tool into a fully interactive educational platform. By bridging this gap, learners could gain deeper insights into the complexity of prehistoric extinctions, making their education more engaging and impactful. Addressing this research gap could greatly enhance the effectiveness of VR in teaching about mass extinction events and environmental change

Aspect	Existing VR Systems	Proposed System
Primary Objective	Enhance engagement and retention in various subjects	Simulate prehistoric extinction events interactively
Interaction Focus	Visualization of historical events	Interactive event triggering (volcano eruptions, meteorites)
Immersion Level	Moderate	High-fidelity immersive experiences
Realism and Accuracy	Limited environmental realism	AI-driven realistic environmental changes
User Engagement	Passive observation.	Active participation and scenario manipulation
Educational Insights	Focus on general education	Specialized knowledge of prehistoric events
Platform Integration	Standalone VR experiences	Integration with educational platforms
User Interface	Limited user control	Comprehensive control with real-time feedback
Accessibility	Requires advanced VR hardware	Cross-platform support for wider accessibility
Simulation Adaptability	Fixed scenarios	Dynamic and adaptive scenario variations
Contribution to Learning	General awareness	In-depth analysis and data visualization
Impact on Understanding	Surface-level knowledge	Experiential understanding of catastrophic events

Table 01. Comparison of Existing VR Systems and Proposed System

1.3 Research Problem

While every day sees a growing effort to include Virtual Reality in educational settings, traditional educational methods cannot express the scale and impact of extinction events through their means. Books, documentaries, and museum exhibitions are good and well, but they tend to be restrictive to static information. They rarely put learners into dynamic environments where they can be involved in those events or interact with them. Their approach also has its limitations with traditional resources, leaving some gap in the learning curve.

Prehistoric extinction events are difficult to explain by conventional teaching. For instance, a textbook may describe how the meteorite impact wiped out the dinosaurs but cannot describe what it was like; the time it took over millions of years and how the environment changed in that time. These methods do not provide opportunities for simulating the impact which these phenomena of climate change, atmospheric shift, and ecosystem development have essentially caused, which is an important core in ascertaining the impact caused by these disastrous happenings. They leave the learners with only one-dimensional thoughts thus restricting their perceptions and abilities in understanding such a historic occurrence of these factors.

Furthermore, if AI were used to simulate the gradual environmental changes brought on by these events, learning would be further deepened. Using AI-driven models, learners would be able to trigger environmental events either earlier or later in time and observe how ecosystems change over time in response. This could include the long-range impact of volcanic ash clouds on climatic conditions or how species changed over time by evolution to adapt to the new environment. Such an approach would also allow users to work out critical thinking by engaging in depth with information. The ability to interact with such simulations would enable learners to investigate various "what-if" scenarios and, importantly, understand the implications of environmental change-which can increase knowledge retention and understanding

The current educational system has not harnessed the potential of such capabilities yet. There is thus a definite and distinct need to develop an interactive VR solution. It is such a platform that should provide users with the facility not only to witness but also to participate in prehistoric extinction events. That would be a great opportunity to experience how such an event took part in shaping the Earth's history and influenced the line of life evolution.

2. Objectives

2.1 Main Objectives

The main objective of the work is to develop an interactive, immersive VR simulation that enables users to experience and understand prehistoric extinction events through a VR environment as an educational tool for exploring the extinction events that shaped Earth's biodiversity. Specific objectives of the study are:

- **Virtual Reality Platform Development:** The simulation should be hosted on an easy-to-use and accessible platform to ensure a smooth and immersive user experience.
- **Interactivity:** Allow the user to interact with the simulation, triggering the extinction event to see what happens to prehistoric life.
- **Ensure Realism and Scientific Accuracy:** Ensuring that events and environments simulated in the game are based on state-of-the-art scientific knowledge about prehistoric Earth and the processes leading to extinction events.

2.2 Specific Objectives

1. **Design Scientifically Correct 3D Environments:**
 - **Simulate Prehistoric Landscapes:** Realistic 3D scenery of prehistoric Earth should be designed, including different landforms like forests, deserts, and oceans.
 - **Reconstruction of Extinction Event:** Correctly model meteorite strikes, and other events causing extinction with real-world scientific data.
 - **Model Climate and Atmospheric Conditions:** Simulate the climatic changes over time due to these catastrophic events
2. **Provide User-Interaction Capabilities to Trigger Extinction Events:**
 - **User-Triggered Events:** Allow users to control and trigger various extinction events at different points in the simulation, experiencing firsthand the devastating effects.
 - **Interactive Learning Features:** Allow manipulation of environmental variables such as temperature, atmospheric pressure, and volcanic activity to understand their influence on extinction outcomes.
 - **Real-Time Feedback** Real-time visual and auditory feedback regarding the results of their interactions, such as environmental changes or changes in species behavior.

2.1 Data Collection

The simulation development is based on a strong dataset. Various scientifically correct and speculative data regarding prehistoric extinction events will be collected with care. It will contain volcanic eruptions, meteorite impacts, climate shifts, and further influences on flora and fauna, forming a comprehensive bedrock for training and testing the AI-driven models in detail. This will enable the dataset to build dynamic, interactive simulations that respond to user interactions in a realistic and educational manner, while it will be grounded in both real-world scientific data and simulated environmental changes.

Data sets:

Global Volcanism Program - <https://volcano.si.edu/>

Earth Impact Database - <https://www.passc.net/EarthImpactDatabase/>

Paleoclimatology Data - <https://www.ncdc.noaa.gov/paleo/>

The Paleobiology Database- <https://paleobiodb.org/>

NASA Astrobiology Institute's Planetary Data System- <https://pds.nasa.gov/>

USGS Earthquake and Volcano Data - <https://volcanoes.usgs.gov/>

National Oceanic and Atmospheric Administration (NOAA) Geophysical Data - <https://www.ngdc.noaa.gov/>

2.2 Process



Figure 02. Process

1. Data Collection

- Gather geological, paleontological, and climate data from reliable sources such as research papers, scientific databases, and museum archives.
- Collect 3D models of prehistoric landscapes, flora, and fauna from open-source platforms or create custom models.
- Include historical records and expert consultations for accuracy.

2. Preprocessing and Data Cleaning

- Standardize and clean the collected data to remove inconsistencies and redundant information.
- Convert different data formats (e.g., 3D models, geological maps, climate records) into a compatible format for VR development.
- Ensure spatial and temporal consistency in datasets.

3. Training Data Preparation

- Prepare datasets for AI-driven elements in the simulation (e.g., behavior of dinosaurs, environmental changes).
- Label and categorize data for machine learning models to enhance interactivity and realism.

4. Feature Engineering

- Develop features such as interactive simulations of extinction events (volcanic eruptions, meteorite impacts).
- Implement dynamic environmental conditions (changing weather, ecosystem evolution).
- Add soundscapes and visual effects to enhance immersion.

5. Virtual Reality Model Development

- Build realistic 3D environments using Unity or Unreal Engine.
- Integrate AI elements to make the simulation adaptive and interactive.
- Optimize performance for different VR platforms.

6. Classification and Simulation Optimization

- Implement algorithms to classify different environmental interactions and their impacts.
- Use real-time feedback mechanisms to optimize the learning experience.

7. Identifying User Engagement and Learning Outcomes

- Assess how users interact with the VR content and track learning outcomes.
- Collect feedback and analytics to refine the content.

8. Evaluation and Risk Assessment

- Conduct user studies to evaluate the effectiveness of the simulation in educational settings.
- Identify potential usability issues and address VR-induced motion sickness.

9. Platform Integration

- Ensure compatibility with educational platforms and institutions.
- Develop support for various VR devices, including standalone and PC-based systems.

10. Impact Analysis and Educational Insights

- Measure the impact of the simulation on knowledge retention and engagement.
- Generate reports and insights to support further improvements and research.

2.3 Requirements

2.3.1 Functional Requirements

Functional requirements describe the cardinal features and functionalities that the VR simulation of prehistoric times must be designed to embed to achieve the desired education for the user.

Core Functionalities:

1. Immersive Prehistoric Environment:
 - Exact 3D landscapes of prehistoric Earth, for example, from the Jurassic or Cretaceous periods.
 - Interactive flora and fauna, as well as geological phenomena.
2. Interactive Simulations:
 - Model extinction events: for example, volcanic eruptions, meteorite impacts, and climate shifts.
 - Allow the user to manipulate variables to make observations.
3. User Navigation and Interaction:
 - Intuitive VR controls for locomotion and manipulation of objects
 - Multi-modal interaction: gesture, voice commands, controllers
4. Educational Content Delivery:
 - Display informational pop-ups, voiceovers, and quizzes on relevant prehistoric events
 - Structured learning paths, depending on different age groups and education levels
5. Real-Time Adaptation and Feedback:

- AI-driven runtime adjustments based on user interactions
 - On-the-spot feedback about learning achievements and exploration so far
6. Data Logging and Analytics:
 - Tracking of user engagement and progress, session time
 - Reports for educators about student performance
 7. Cross-Platform Compatibility:
 - Support for VR headsets like Oculus, HTC Vive, and desktop VR experiences.
 - Mobile compatibility for simplified versions.

2.3.2 Non-Functional Requirements

Non-functional requirements are the quality attributes and constraints of the system that ensure the efficiency, reliability, and usability of the system.

1. Performance:
 - The VR simulation should run above 90 FPS to avoid motion sickness.
 - Low latency of less than 20ms for interactions to make the experience smooth.
2. Scalability:
 - The system shall be capable of running up to 50 concurrent users in multiplayer mode without performance degradation.
3. Usability:
 - Intuitive UI/UX optimized for educational purposes, with a minimal learning curve.
 - Multiple control modes, such as voice commands and gesture control, for accessibility.
4. Security:
 - User data can be stored and transmitted securely; GDPR compliant.
 - Role-based access control to avoid unauthorized modification of education content.
5. Compatibility:
 - Compatible with a wide variety of VR headsets and hardware configurations.
 - Integration with existing Learning Management Systems.
6. Maintainability
 - Architecture that would easily allow patch updates and addition of new content.
 - Regular updates and patches should be done without any downtime.
7. Reliability

- The system should allow for 99.9% uptime, which means the system is available to students and educators.
- Error handling and logging mechanisms should be implemented to debug and perform improvements upon the system.

2.3.3 Hardware Requirements

1. VR Headsets with built in tracking
2. Development of Workstation with CPU,GPU, RAM, Storage, Monitor

2.3.4 Software Requirements

1. VR development Platforms
 - Unity 3D
 - Blender
2. Programming Languages
 - C#
 - Python
3. 3D Modelling Tools
 - Blender
 - Substance painter
4. AI data processing
 - TensorFlow
 - OpenCV
5. Database Options
 - Firebase
6. Cloud Services
 - Google Drive
7. Version Control:
 - GitHub
8. Deployment Platforms
 - Meta Quest Store

2.4 Testing

Testing is a very crucial stage that remains for the verification of functionality, reliability, and overall performance of the VR simulation. Various testing methodologies shall be used to validate various aspects within the system.

2.4.1 Unit Testing

Here, each and every component of the VR simulation is independently tested as a stand-alone module so that it produces expected results.

Scope:

- VR interactions include object manipulation and movement tracking.
- AI-induced environmental changes such as volcanic eruptions and meteorite impacts.
- 3D model rendering and animation performance.
- Input devices include VR controllers, sensors, and so on.

Tools Used:

- Unity Test Framework: for Unity-based development,
- NUnit: for C# scripting test,
- PyTest: for AI-related Python modules.

Outcome:

- Every module should have been tested in isolation and then integrated to form the final system.

2.4.2 Integration Testing

The aim is to perform an integration test on the interaction across different modules, focusing on data flow and communication among system parts.

Scope:

- VR engine Unity/Unreal with AI model integration.
- Interaction of user input to environmental responses.
- Database interaction - Firebase/SQLite
- Cross-platform - VR headset, Web-based platforms.

Tools Used:

- Unity Play Mode Tests
- Postman for testing in Cloud-based APIs
- Firebase Emulator for Database Testing

Expected Outcome:

- That there is smooth interaction and data exchange between system parts.
-

2.4.3 System Testing

The objective will be to carry out the validation of the complete VR simulation system in real conditions regarding the fulfillment of functional and non-functional requirements.

Scope:

- Performance testing: Frame rate, load times
- User experience testing: Ease of interaction, educative value.
- Compatibility testing on different VR hardware: Meta Quest and HTC Vive.
- Stress testing: At various environmental conditions.

Tools Used:

- Unity Profiler: to assess performance metrics
- Oculus Debug Tool: for performance monitoring of VR
- Surveys and user feedback collection.

Expected Outcome:

- A fully functional, immersive VR time-travel simulation back to the prehistoric era for education and fun.

2.5 Gantt Chart

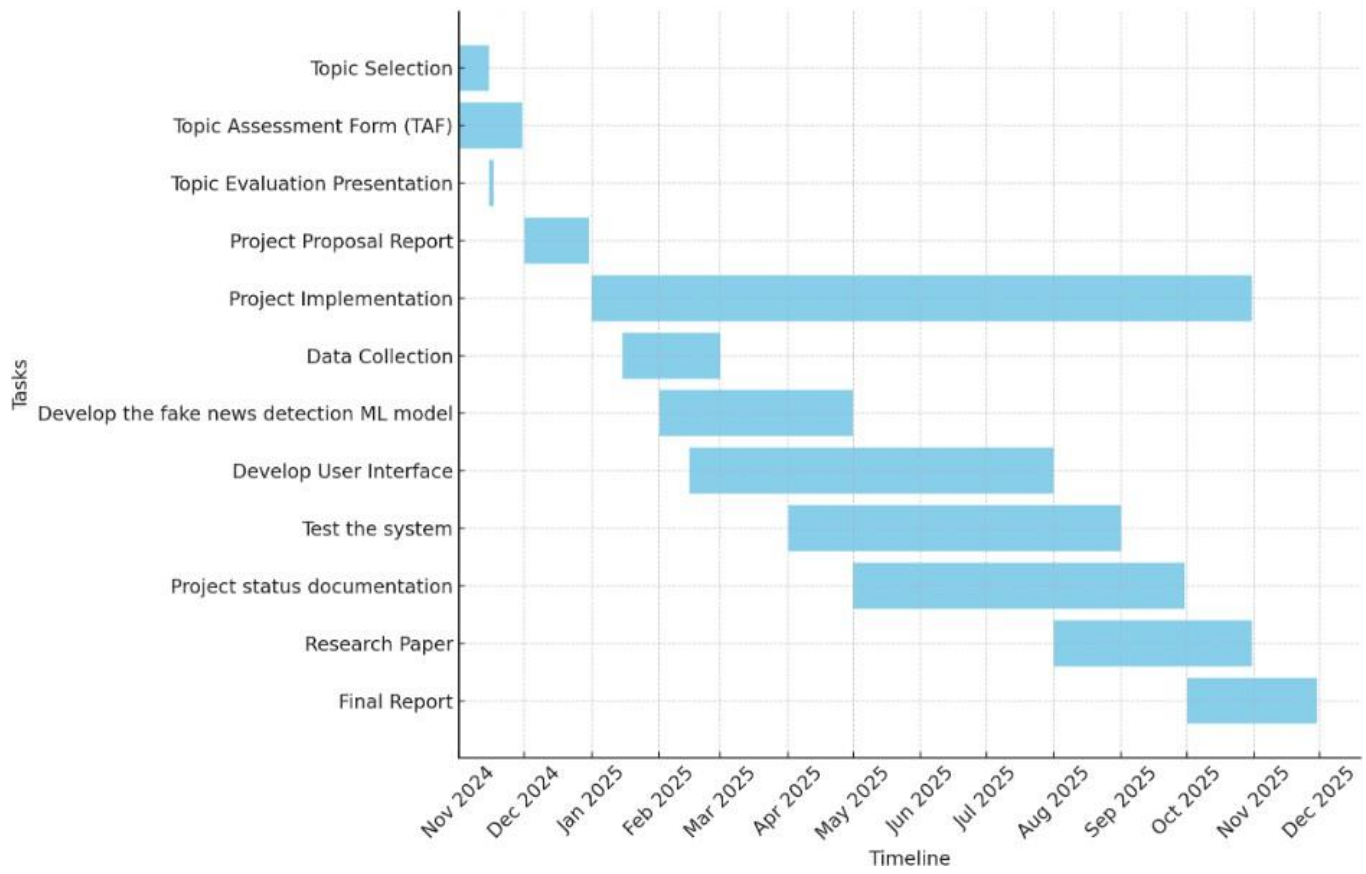


Table 02. Gantt Chart for the Research

3. Estimated Budget

Task	Cost in LKR
VR headset Rental	15, 000
Miscellaneous (Adapters, Cables, Repairs)	5,000
Testing & Deployment	10, 000
Total	30, 000

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